

PyHydroGeophysX: An Extensible Open-Source Platform for Integrating Hydrological Models with Geophysical Measurements

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Abstract

Hydrological models and geophysical measurements are important tools for understanding subsurface hydrological processes essential for water resource management, yet they typically remain disconnected due to technical barriers. We present PyHydroGeophysX, an open-source Python platform bridging this gap by providing standardized interfaces between hydrological modeling software (MODFLOW, ParFlow) and geophysical simulation tools (PyGIMLi, SimPEG). The platform implements bidirectional workflows: translating hydrological outputs into simulated geophysical responses through petrophysical models, and extracting hydrological information from geophysical inversions. Key features include modular petrophysical relationships, parallel computing, 3D mesh creation, and mesh interpolation. The modular architecture of PyHydroGeophysX enables researchers to incorporate additional models and methods, fostering broader adoption of integrated hydrogeophysical approaches.

Keywords: Hydrogeophysics, Python, Hydrological modeling, Open-source, Inversion, Petrophysics

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Code metadata

Nr.	Code metadata description	Metadata
C1	Current code version	v1.0
C2	Permanent link to code/repository used for this code version	https://github.com/geohang/PyHydroGeophysX
C3	Permanent link to Reproducible Capsule	https://doi.org/10.5281/zenodo.17025139
C4	Legal Code License	Apache License 2.0
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python 3.8+
C7	Compilation requirements, operating environments & dependencies	NumPy, SciPy, Matplotlib, Flopy, PyGIMLi, SimPEG, Pftools; Linux/macOS/Windows
C8	If available Link to developer documentation/manual	https://geohang.github.io/PyHydroGeophysX/
C9	Support email for questions	Hangchen.work@gmail.com

1. Motivation and significance

Understanding hydrological processes is essential for agriculture, aquifer management, ecohydrology, and water resources in mountainous watersheds [3]. Subsurface hydrology is especially critical, as it regulates water availability and recharge, sustaining ecosystems and water resources [10]. However, our understanding of subsurface water dynamics remains limited, particularly in heterogeneous environments [13]. Field measurements of soil moisture and water levels provide valuable insights into subsurface hydrologic processes [23], but such point observations often fail to capture the spatial variability of subsurface properties [26].

Two complementary approaches have emerged to address this challenge. Process-based hydrological models, such as MODFLOW [18] and ParFlow [17], simulate water flow and provide predictions where direct measurements are unavailable [25]. However, these models require spatially distributed subsurface parameters that are often poorly constrained by sparse observations [21]. Geophysical methods, including electrical resistivity tomography (ERT), seismic refraction tomography (SRT), and electromagnetics (EM), provide spatial information about subsurface structure and properties [32, 24, 11, 34]. However, geophysical approaches face limitations, including the fact that they provide indirect measurements that require petrophysical relationships to translate geophysical attributes into hydrological parameters

[33, 12]. These methods also offer limited process understanding without complementary modeling [9].

The integration of hydrological models and geophysical measurements represents a powerful paradigm for advancing hydrological science [29]. Process-based hydrological models provide a physical context for interpreting geophysical observations, helping to distinguish between competing hypotheses about subsurface behavior and hydrological processes [30]. The coupled inversion of hydrological and geophysical data has shown potential to reduce uncertainty in both domains [14, 28]. Geophysical measurements can also improve hydrological models through improved spatial characterization of the subsurface [19]. For example, Chen et al. demonstrated the integration of seismic refraction constraints into hydrological models to improve subsurface parameterization [6, 5], while Perzan et al. and Uhlemann et al. used resistivity methods to parameterize hydrological models across diverse groundwater settings [27, 35].

Despite the potential for integration, researchers in these domains often work in isolation using specialized software with incompatible data formats and workflows. While some integrated tools exist, such as PFLOTRAN-E4D [16], they remain confined to specialized software and limit broader adoption of hydrogeophysical methods. General-purpose parameter estimation tools like PEST++ [37] primarily support hydrological model calibration and do not natively incorporate geophysical forward modeling or data assimilation.

To address this gap, we present PyHydroGeophysX, an open-source platform that integrates geophysical measurements with hydrological models in a unified Python ecosystem. The platform builds on PyGIMLi [31] and SimPEG [8] for geophysical modeling while providing standardized interfaces to MODFLOW and ParFlow [17, 18]. PyHydroGeophysX establishes an iterative feedback loop between model simulations and geophysical measurements, where models guide survey design and interpretation while measurements improve model parameterization. By providing pre-built petrophysical modules, mesh interpolation utilities, and workflow templates, PyHydroGeophysX reduces the technical barriers for researchers seeking to combine these approaches.

2. Software description

2.1. Software architecture

PyHydroGeophysX adopts a modular, layered architecture designed to support bidirectional hydrogeophysical coupling workflows (Figure 1). The modular architecture supports flexible workflows in both directions: con-

verting hydrological model outputs to predicted geophysical responses and extracting hydrological parameters from geophysical measurements.

The Data I/O layer standardizes interactions with external software. On the hydrological side, dedicated interfaces built on FloPy and ParFlow utilities access outputs from MODFLOW 6 and ParFlow simulations [18, 17, 15]. On the geophysical side, RESIPy-compatible workflows enable consistent data loading, quality control, and export for electrical survey data [2].

The Core processing layer provides shared computational infrastructure used throughout the platform. This includes 2D and 3D mesh utilities (e.g., **Mesh3DCreator**), interpolation between discretizations, and solver backends (including CGLS and optional GPU acceleration). A dedicated petrophysics module translates hydrological state variables into geophysical property models, ensuring that physical property transformations remain transparent and configurable.

The Geophysics processing layer integrates forward modeling and inversion engines using community-supported frameworks such as PyGIMLi [31] and SimPEG [8]. Forward operators support ERT, SRT, and EM workflows. Inversion modules include single-time and time-lapse approaches, with temporal regularization and parallel execution capabilities for monitoring-scale datasets.

The Integration layer exposes high-level workflow modules that connect hydrological and geophysical components in both directions. The HM to GM pathway translates hydrological states into geophysical responses, while the GM to HM pathway incorporates inversion results into hydrologic model parameterization. This architecture reflects the iterative ModEx paradigm advocated in hydrogeophysics and demonstrated in catchment-scale applications [6, 7].

2.2. Software functionalities

PyHydroGeophysX provides end-to-end hydrogeophysical workflow support through several key functional modules. Hydrological model interfaces allow extraction of spatial fields and time series from MODFLOW 6 and ParFlow simulations [18, 17]. Classes such as **MODFLOWWaterContent** and **ParflowSaturation** provide consistent access to water content, saturation, and porosity outputs across different simulator backends.

Petrophysical transformations convert hydrological state variables into geophysical property models. For electrical resistivity, the platform implements Archie-type and Waxman–Smits formulations [1, 36]. For seismic velocities, Hertz–Mindlin and differential effective medium approaches are available [22]. Users can configure layer-specific relationships or implement custom petrophysical models.

Forward modeling capabilities include: (1) ERT forward modeling in 2D and 3D using finite-element formulations via PyGIMLi [31], (2) SRT forward modeling based on travel-time solutions, and (3) EM/TDEM forward responses using SimPEG [8] with finite volume methods.

Inversion capabilities include single-time inversion, time-lapse inversion with temporal regularization (ℓ_1 or ℓ_2), windowed time-lapse inversion (`WindowedTimeLapseERTInversion`) for large monitoring datasets [38], and structure-constrained inversion using seismic interfaces [4]. Time-lapse inversion strategies follow principles established in hydrogeophysical monitoring studies [20].

High-level convenience functions encode common coupling patterns to streamline workflows. For example, `hydro_to_ert()` and `hydro_to_srt()` translate hydrological state variables into geophysical parameter models and execute forward simulations in a single workflow step. Additional utilities such as `extract_velocity_structure()` and `krige_seismic_velocity_3d()` support structural information transfer between seismic tomography products and hydrological model domains. These functionalities formalize workflows previously implemented in case-study research [4, 6] into a reproducible and extensible software framework.

3. Illustrative Examples

We demonstrate PyHydroGeophysX using a synthetic headwater catchment based on the Dry Creek Experimental Watershed, Idaho [23]. This example illustrates both directions of the hydrogeophysical workflow: translating hydrological model outputs into geophysical responses, and extracting structural information from geophysical data for hydrological model parameterization.

3.1. Hydrological Model Setup

The synthetic model represents a small mountainous catchment (~ 0.02 km²) with elevations ranging from 1560 to 1650 m. The subsurface comprises three geological layers: regolith, fractured bedrock, and fresh bedrock (Figures 2a,b). Porosity decreases with depth from 0.45 at the surface to 0.05 in fresh bedrock (Figure 2c), with hydraulic conductivity derived from the Kozeny–Carman equation. MODFLOW 6 simulates water dynamics over a complete water year using field-derived precipitation and evapotranspiration data.

The `ProfileInterpolator` class extracts properties along a 110 m transect crossing the central valley (Figure 2a). Figure 2c shows the extracted

porosity distribution, while Figure 2d displays the saturation field from MODFLOW, revealing the characteristic transition from unsaturated conditions near the surface to full saturation below the water table.

3.2. *Petrophysical Transformations*

PyHydroGeophysX implements layer-specific petrophysical models to convert hydrological properties into geophysical parameters.

Electrical resistivity (Figure 3): The Waxman–Smits model is applied for the clay-rich regolith layer, while Archie’s law governs the fractured bedrock. Figure 3a shows the resulting vertical resistivity profile, with high values in the shallow unsaturated zone decreasing with depth. Figures 3b and 3c demonstrate the porosity–resistivity relationships for each layer, highlighting that resistivity is strongly sensitive to saturation variations—making ERT effective for monitoring hydrological dynamics but challenging for structural characterization.

Seismic velocity (Figure 4): The Hertz–Mindlin model is applied for granular regolith, while the differential effective medium (DEM) model governs bedrock layers. Figure 4a shows distinct velocity stratification with clear boundaries between geological units. Figures 4b and 4c demonstrate that velocity correlates primarily with porosity rather than saturation, making seismic methods particularly valuable for delineating subsurface structure.

3.3. *Time-Lapse ERT Forward Modeling*

Figure 5 demonstrates the framework’s capability for simulating time-lapse geophysical monitoring. Daily precipitation drives the hydrological model (Figure 5a), and the resulting water content variations are translated into apparent resistivity measurements using parallel forward modeling via `joblib.Parallel`.

The apparent resistivity evolution (Figure 5b) shows clear decreases following major precipitation events, particularly after day 200. Water content and resistivity cross-sections at selected time points (Figures 5c,d) illustrate the infiltration dynamics: Day 30 shows dry conditions with high resistivity; Day 210 displays maximum water content after sustained precipitation with corresponding resistivity decreases to $\sim 400 \text{ } \Omega\cdot\text{m}$; Day 330 shows intermediate recovery conditions. These simulations support survey optimization by identifying that valley locations (50–60 m) and mid-depth electrode configurations (#200–400) show the strongest sensitivity to hydrological changes.

3.4. *Seismic Tomography and Structure Extraction*

The inverse workflow is demonstrated using synthetic SRT data (Figure 6). Two array configurations are compared: a short array (72 geophones

over 72 m) and a long array (90 geophones over 110 m). Figure 6a shows the simulated travel-time data for both configurations.

The inverted velocity models (Figure 6b) demonstrate how array design influences structural resolution. The short array provides higher near-surface resolution, clearly identifying the regolith layer. The long array achieves greater depth penetration (>40 m), revealing all three geological units: regolith, fractured bedrock, and fresh bedrock. Black lines indicate the layer boundaries extracted using `extract_velocity_structure()`, which can be directly incorporated into hydrological model domains using `krige_seismic_velocity_3d()` for 3D interpolation.

4. Impact

PyHydroGeophysX is built around the ModEx paradigm, which enables iterative feedback between hydrological models and geophysical measurements (Figure 7). The framework addresses a recognized gap between two complementary approaches to subsurface characterization: hydrological models provide process-based physics, predictive capabilities, and scenario testing, but face challenges with parameter uncertainty, simplified heterogeneity representation, and calibration constraints; geophysical measurements offer spatial imaging, non-invasive monitoring, and heterogeneity detection, but are limited by their indirect nature, constrained process understanding, and inversion non-uniqueness. PyHydroGeophysX bridges these limitations through bidirectional information exchange. In the HM to GM direction, hydrological models guide survey design optimization, provide physical explanations for observed geophysical signals, and constrain inversions with process-based information. In the GM to HM direction, geophysical data identify spatial structures, extract distributed hydrological properties, and provide constraints for model calibration. For example, seismic tomography can delineate subsurface layer boundaries to improve heterogeneity representation in hydrological models, and the refined models can then identify optimal locations for resistivity monitoring, ensuring surveys target the dynamics most relevant to hydrological processes. This continuous feedback enables new research directions including: (1) systematic evaluation of geophysical survey sensitivity to hydrological processes, supporting optimized monitoring design; (2) multi-physics characterization combining ERT, SRT, and EM for comprehensive subsurface imaging; and (3) uncertainty-aware parameter estimation through Monte Carlo methods.

The PyHydroGeophysX framework was designed with extensibility as a core principle, enabling straightforward integration of additional hydrological models and geophysical methods beyond those currently implemented.

This flexibility stems from the modular architecture shown in Figure 1, where clearly defined interfaces separate core functionality from specific implementation details. New hydrological model support requires only implementing data loader classes that extract relevant properties (e.g., saturation and porosity) into standardized internal formats. For example, adding support for the Advanced Terrestrial Simulator would follow the pattern established by the existing MODFLOW and ParFlow interfaces. Geophysical method extensibility follows a parallel approach. Since the framework already transforms hydrological outputs into physical property distributions (e.g., resistivity, velocity), adding new geophysical methods primarily involves linking appropriate forward modeling functions. Integration with PyGIMLi provides access to additional methods such as gravity and induced polarization, while connections to SimPEG extend capabilities to airborne electromagnetics and magnetotellurics.

5. Conclusions

In this paper, we have presented PyHydroGeophysX, an open-source platform that bridges the gap between geophysical measurements and hydrological models within an integrated, extensible framework. This platform addresses critical challenges in subsurface hydrologic science by enabling bidirectional workflows that use the complementary strengths of both disciplines while mitigating their individual limitations. Through the implementation of a unified framework, advanced time-lapse inversion techniques, petrophysical models, and flexible data structures, PyHydroGeophysX transforms traditional approaches into a unified ecosystem for comprehensive subsurface characterization. Our synthetic catchment demonstration illustrated the framework’s capabilities for both forward modeling: predicting geophysical signatures from hydrological processes, and inverse application: extracting hydrologically relevant parameters from geophysical measurements. Looking forward, PyHydroGeophysX represents a foundation for advancing integrated approaches to hydrological science that will enable more robust subsurface characterization of hydrological processes across multiple scales. By lowering technical barriers to implementation through modular design, comprehensive documentation, and simplified workflows, we aim to foster broader adoption of hydrogeophysical methods among researchers and practitioners from diverse backgrounds. The open-source nature of the platform creates opportunities for community-driven extensions that will continuously improve its capabilities and applications. As growing water demands intensify pressure on water resources worldwide, especially in vulnerable mountainous watersheds, tools like PyHydroGeophysX will become increasingly valuable for

providing the detailed subsurface information needed to support sustainable management strategies.

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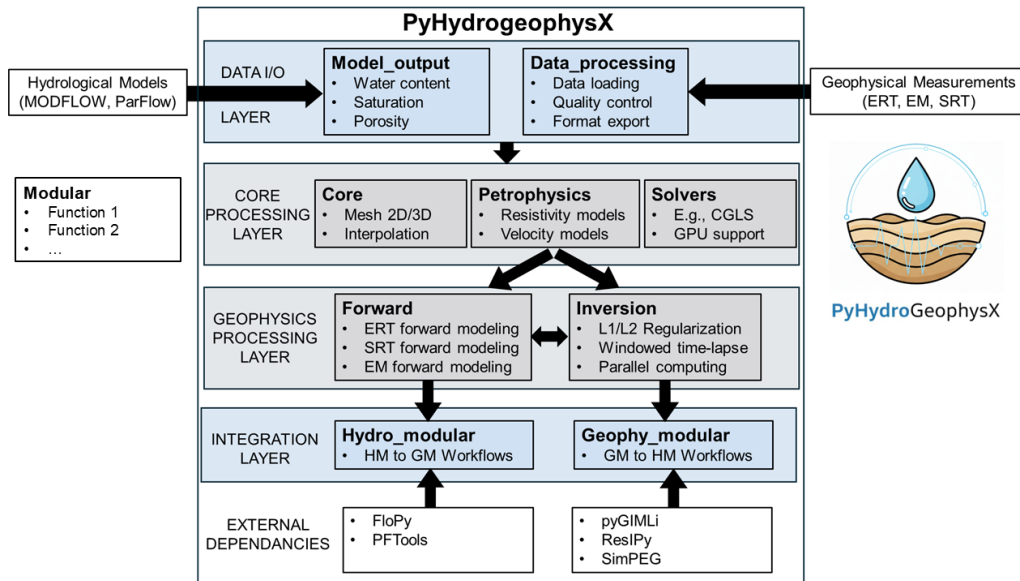


Figure 1: PyHydroGeophysX architecture and coupled hydrogeophysical workflow. The platform comprises four layers: Data I/O, Core processing, Geophysics processing, and Integration. Arrows indicate information flow and iterative feedback between models and measurements.

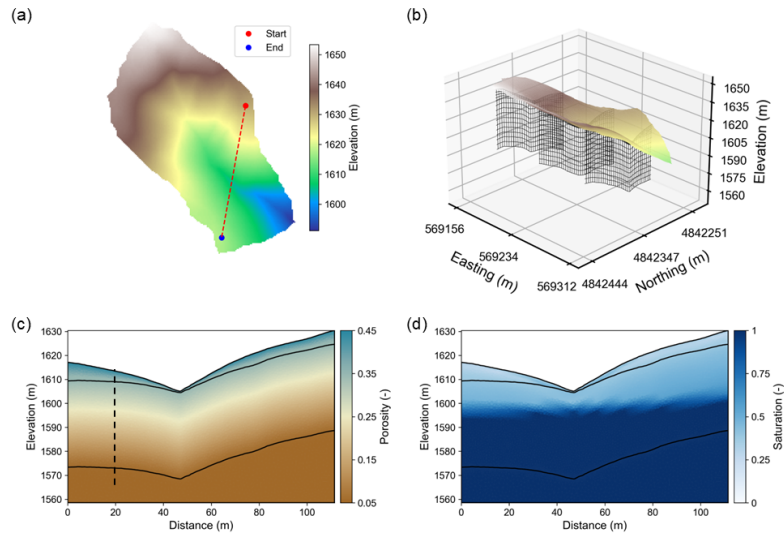


Figure 2: MODFLOW hydrological model setup and subsurface properties. (a) Plan view of catchment elevation with survey transect indicated by the red dashed line from Start (red dot) to End (blue dot). (b) Three-dimensional computational mesh for MODFLOW 6 simulation. (c) Porosity distribution along the transect showing three-layer structure: regolith, fractured bedrock, and fresh bedrock. Black lines indicate layer boundaries. (d) Water saturation from MODFLOW simulation showing unsaturated zone transitioning to full saturation at depth.

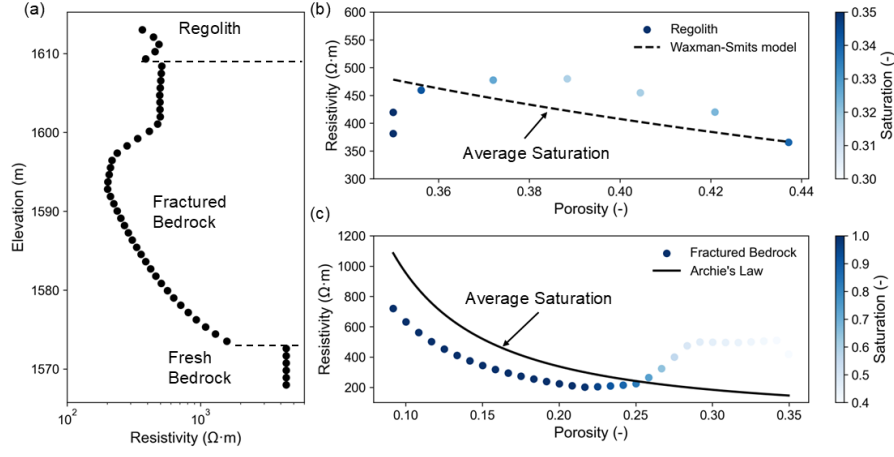


Figure 3: Electrical resistivity model and petrophysical relationships at the 20 m profile location. (a) Vertical resistivity profile showing distinct geological layers with boundaries indicated by dashed lines. (b) Waxman–Smits model for regolith layer relating resistivity to porosity; color indicates saturation. The dashed line represents model predictions based on average saturation. (c) Archie’s law for fractured bedrock; the solid line represents model predictions, with scatter indicating strong saturation control on resistivity.

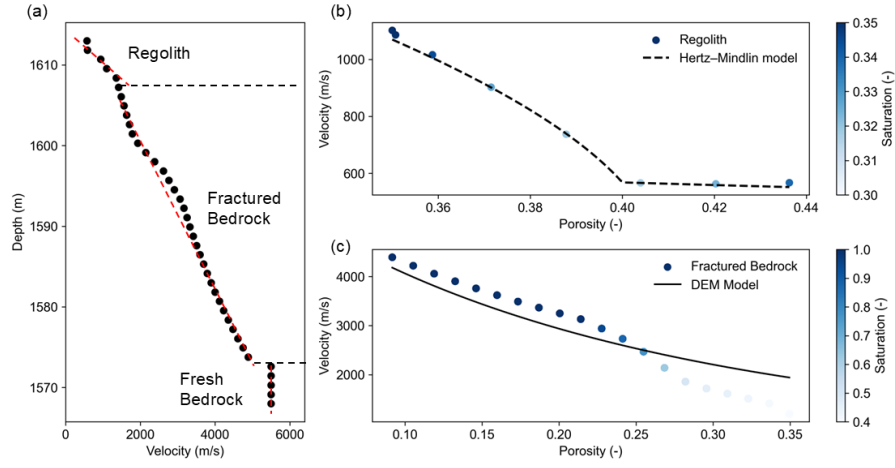


Figure 4: Seismic velocity model and petrophysical relationships at the 20 m profile location. (a) Vertical velocity profile with clear structural boundaries between geological layers indicated by dashed lines. (b) Hertz–Mindlin model for regolith showing velocity–porosity relationship; data cluster closely around the model curve, indicating velocity is mainly controlled by porosity. (c) DEM model for fractured bedrock; velocity strongly correlates with porosity with minimal scatter.

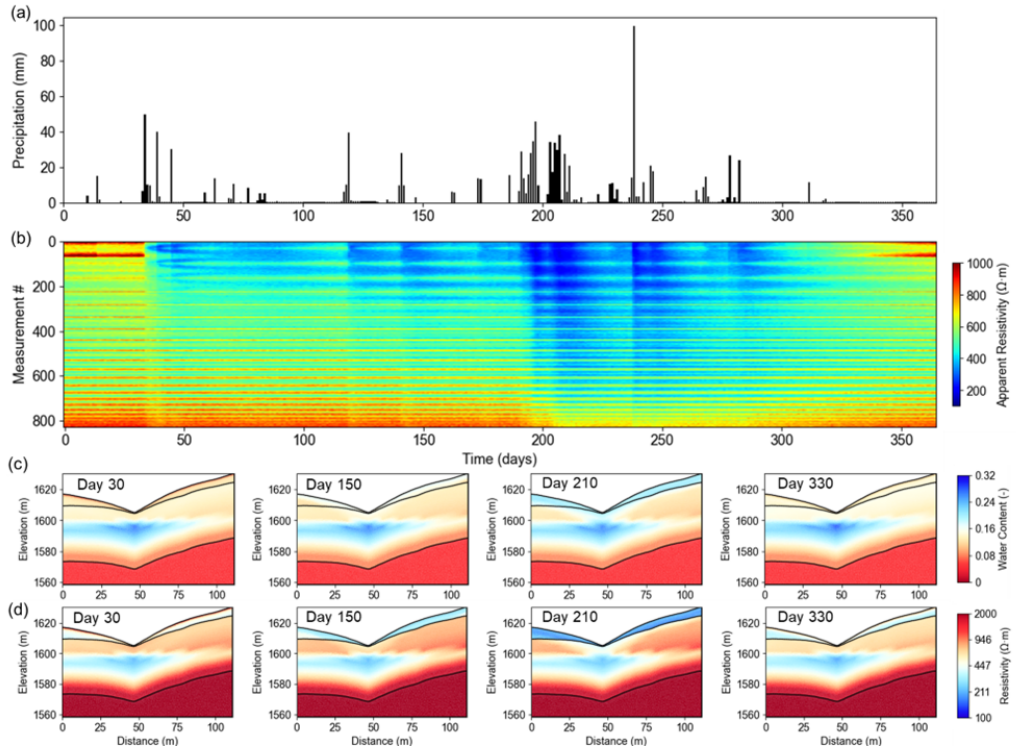


Figure 5: Time-lapse hydrological and geophysical monitoring over a water year. (a) Daily precipitation used as input for the hydrological model. (b) Simulated time-lapse apparent resistivity measurements, showing temporal variations in response to precipitation events. (c) Water content cross-sections at selected time points (Days 30, 150, 210, and 330). (d) Corresponding electrical resistivity cross-sections at the same time points.

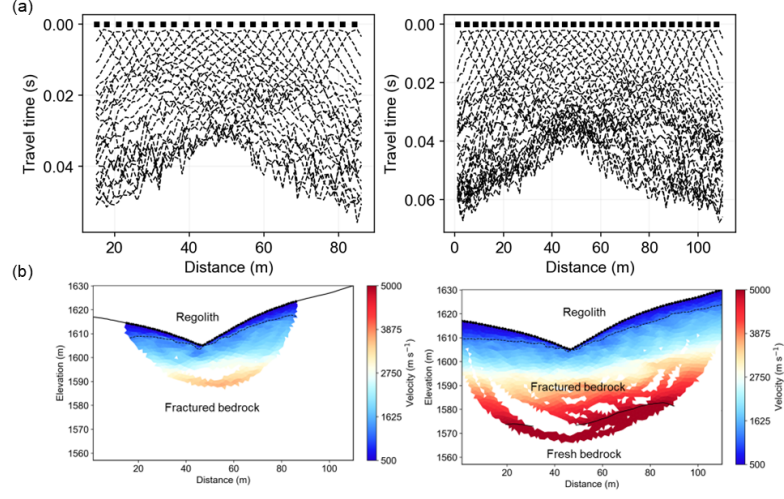


Figure 6: Seismic refraction tomography for subsurface structure extraction. (a) Simulated travel-time data for short array (72 geophones over 72 m, left) and long array (90 geophones over 110 m, right) configurations. (b) Inverted velocity models showing subsurface structure. Short array (left) provides higher near-surface resolution that identifies the regolith layer. Long array (right) achieves greater depth penetration, revealing all three geologic units: regolith, fractured bedrock, and fresh bedrock. Black lines indicate layer boundaries that would be extracted for hydrological models.

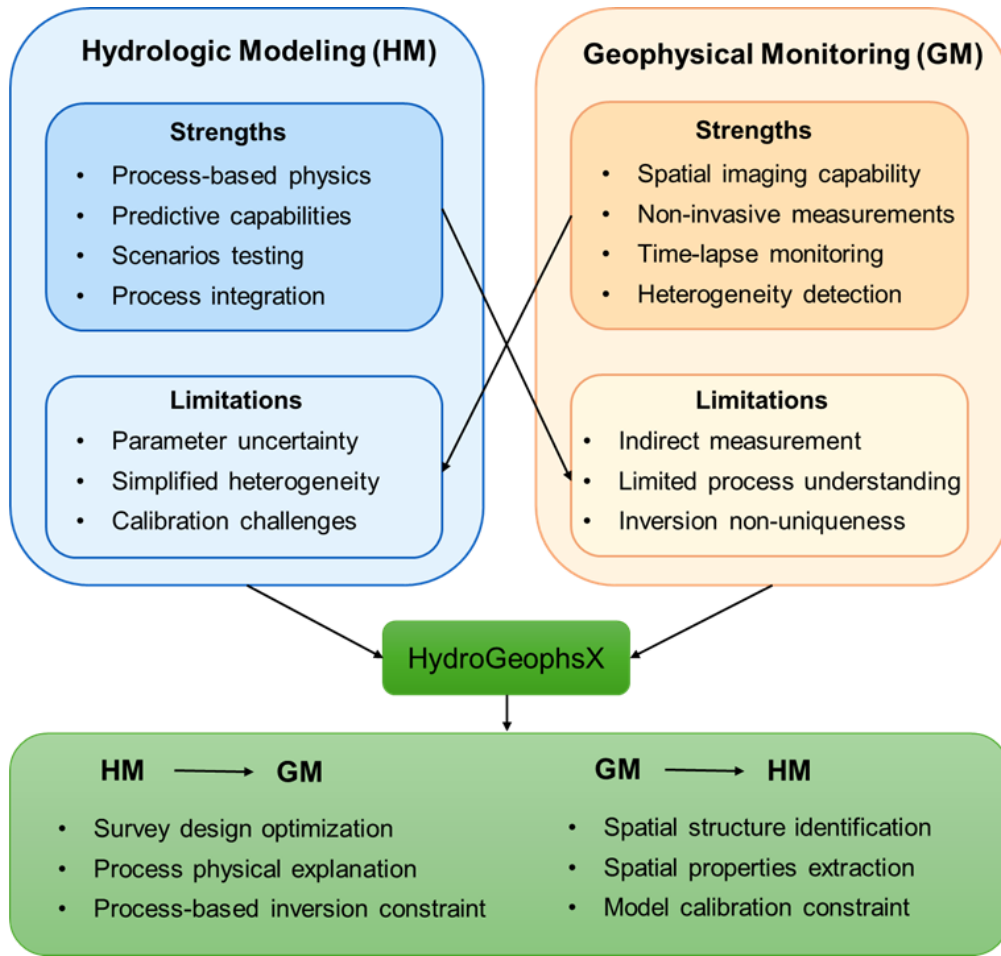


Figure 7: Conceptual framework of PyHydroGeophysX showing the integration of Hydrological Models (HM) and Geophysical Measurements (GM). The diagram highlights the complementary strengths and limitations of each approach and illustrates bidirectional information exchange enabled by PyHydroGeophysX, combining strengths from both disciplines to overcome their individual limitations.